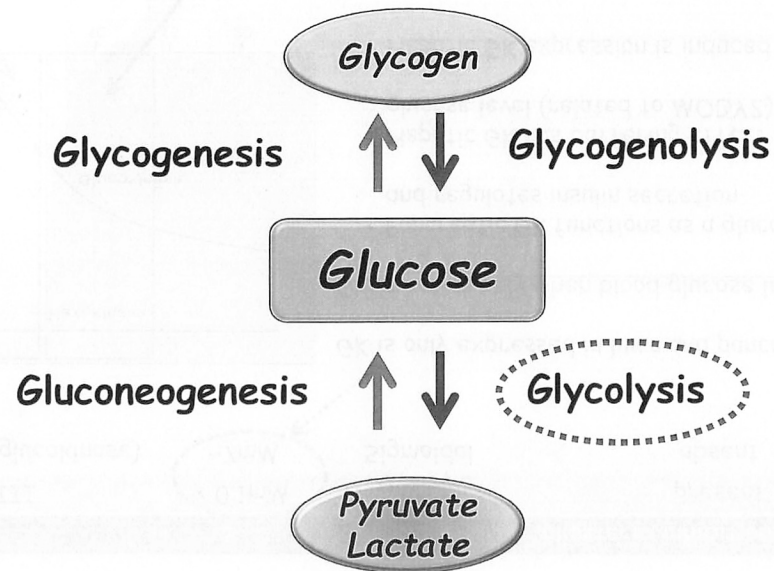
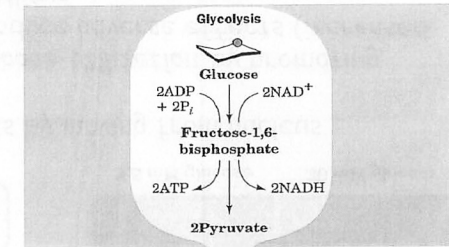


Glucose is a central molecule of carbohydrate metabolism



Glycolysis

- consists of 10 reactions
- occurs in all cells of human body
- occurs in cytosol even under anaerobic condition (oxygen is not needed)
- produces 2mol of ATP and NADH from 1mol of glucose
- ends with 2mol of pyruvate, which is either
 - oxidized completely to CO_2 and H_2O in the presence of oxygen
 - or
 - metabolized to lactate under anaerobic conditions
- In certain cells such as *muscle cells*, it serves as a sole energy-producing system

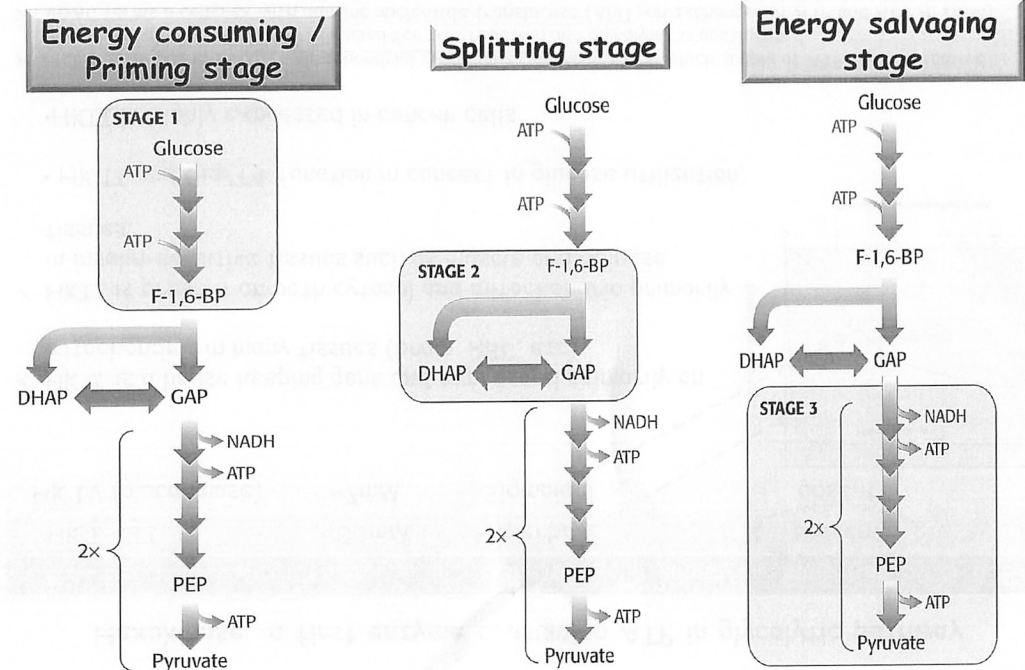


Reactions of Glycolysis

Step	Reaction	Enzyme
1	$\text{Glucose} + \text{ATP} \rightarrow \text{glucose 6-phosphate} + \text{ADP} + \text{H}^+$	Hexokinase
2	$\text{Glucose 6-phosphate} \rightleftharpoons \text{fructose 6-phosphate}$	Phosphoglucose isomerase (Glucose-6-Phosphate Isomerase)
3	$\text{Fructose 6-phosphate} + \text{ATP} \rightarrow \text{fructose 1,6-bisphosphate} + \text{ADP} + \text{H}^+$	Phosphofructokinase
4	$\text{Fructose 1,6-bisphosphate} \rightleftharpoons \text{dihydroxyacetonephosphate} + \text{glyceraldehyde 3-phosphate}$	Aldolase
5	$\text{Dihydroxyacetone phosphate} \rightleftharpoons \text{glyceraldehyde 3-phosphate}$	Triose phosphate isomerase
6	$\text{Glyceraldehyde 3-phosphate} + \text{P}_i + \text{NAD}^+ \rightleftharpoons \text{1,3-bisphosphoglycerate} + \text{NADH} + \text{H}^+$	Glyceraldehyde 3-phosphate dehydrogenase
7	$\text{1,3-Bisphosphoglycerate} + \text{ADP} \rightleftharpoons \text{3-phosphoglycerate} + \text{ATP}$	Phosphoglycerate kinase
8	$\text{3-Phosphoglycerate} \rightleftharpoons \text{2-phosphoglycerate}$	Phosphoglycerate mutase
9	$\text{2-Phosphoglycerate} \rightleftharpoons \text{phosphoenolpyruvate} + \text{H}_2\text{O}$	Enolase
10	$\text{Phosphoenolpyruvate} + \text{ADP} + \text{H}^+ \rightarrow \text{pyruvate} + \text{ATP}$	Pyruvate kinase

Three irreversible steps

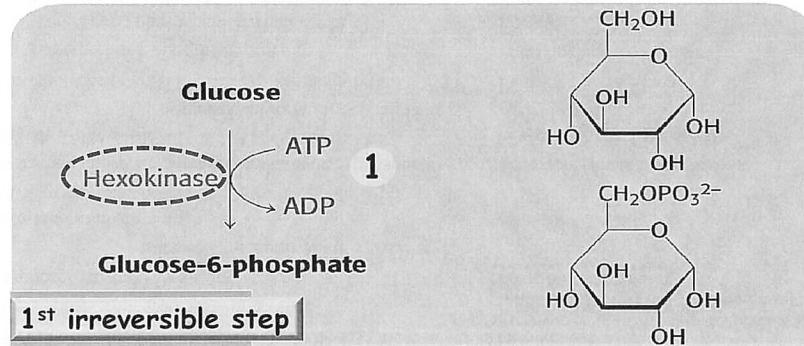
Glycolysis occurs in three stages



Priming stage uses 2mol of ATP to convert glucose to F-1,6-BP

Glucose is trapped as G-6-P in the cell

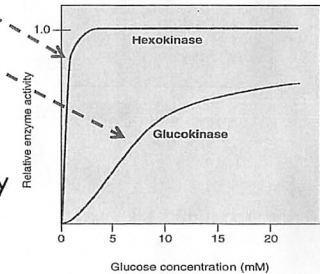
Stage 1



Hexokinase, a first enzyme consuming ATP in glycolytic pathway

	$S_{0.5}$	Kinetics	allosteric inhibition by G-6-P
HK I~III	< 0.1mM	Hyperbolic	present
HK IV (glucokinase)	~7mM	Sigmoidal	absent

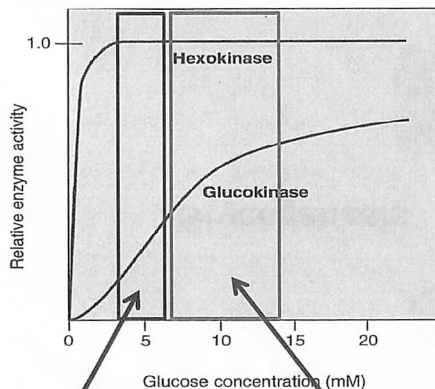
- ✓ HK I is a house keeping gene and expressed primarily on mitochondria in many tissues (brain, RBC, etc) .
- ✓ HKII is present on both cytosol and mitochondria primarily in insulin-sensitive tissues such as muscle and adipose tissues.
- HK II and GLUT4 function in concert in glucose utilization.
- HKII is highly expressed in cancer cells



- HKI & II is bound to outer mitochondrial membrane (via VDACs), by which supply of ATP from oxidative phosphorylation is preferentially used for this reaction and glycolysis is accelerated.
- VDAC forms a complex with adenine nucleotide translocase (ANT, an exchanger of ATP and ADP in IMM).
- Binding of HK to VDAC also stimulates ANT activity.
- Insulin stimulates binding of HKII to VDAC, supporting efficient glucose uptake upon food intake.

Hexokinase, a first enzyme consuming ATP in glycolytic pathway

	$S_{0.5}$	Kinetics	allosteric inhibition by G-6-P
HK I~III	< 0.1mM	Hyperbolic	present
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GK is only expressed in liver and pancreatic β -cells.

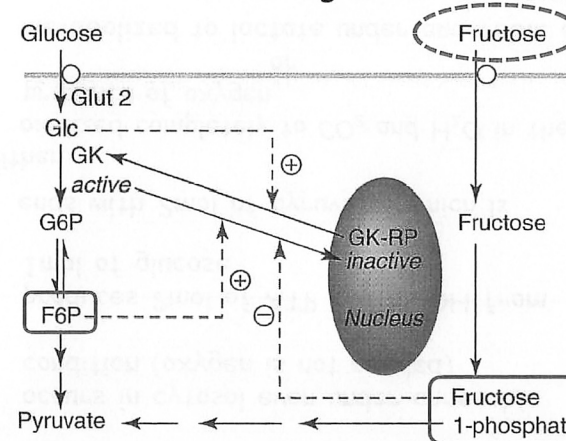
GK is active only when blood glucose levels are high

- Pancreatic GK functions as a glucose sensor and regulates insulin secretion
- Hepatic GK has buffering effect on blood glucose level (related to MODY2)
- Hepatic GK expression is induced by insulin

Normal BS levels

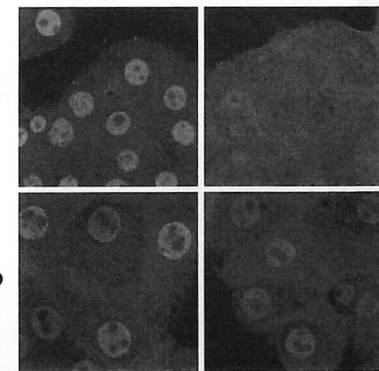
Postprandial BS levels

Role of nuclear GK-RP (glucokinase regulatory protein) in the regulation of localizing GK in nucleus



GK

GKRP



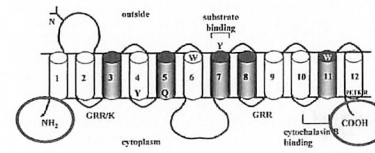
5.5 mM glucose 30 mM glucose

- GK is active only when glucose enters the cells by moving from nucleus
- High fructose intakes stimulate hepatic glucose utilization by promoting dissociation of GK from GK-RP, which may produce adverse effects (increased lipogenesis and high TGmenia) in *diabetes mellitus*

Glucose Transporter

1. Facilitated Transporter: **GLUT (Glucose transporter)**
2. Active Transporter: **Na⁺/Glucose co-transporter (Sodium/glucose cotransporter; SGLT)**:

Facilitated Transporter: **GLUT (glucose transporter): 14 human GLUT proteins**



- All GLUT has 12 transmembrane segments.
- GLUT acts as bidirectional transport of glucose depending on the concentration gradient.
- Rate of glucose transport by GLUT is much faster than that of metabolizing glucose.

➔ GLUT is not rate-limiting step for glucose metabolism

Name	Tissue location	K_m	Comments
GLUT1	All mammalian tissues RBC, BBB, Placenta	1 mM	Basal glucose uptake *1
GLUT2	Liver and pancreatic β cells Reduction of islet GLUT2 in glucolipotoxicity Basolateral Mem in intestine and kidney PCT	15–20 mM High capacity/low affinity	In the pancreas, plays a role in regulation of insulin In the liver, removes excess glucose from the blood
GLUT3	All mammalian tissues	1 mM	Basal glucose uptake
GLUT4	Muscle and fat cells Heart	5 mM	Amount in muscle plasma membrane increases with endurance training
GLUT5	Small intestine Apical Mem, Increase by dietary fructose		Primarily a fructose transporter

Normal glucose levels: 75-105 mg/dl (3.9-5.8mM)

Glut4 is translocated to plasma membrane in response to insulin in skeletal muscles and adipose tissues

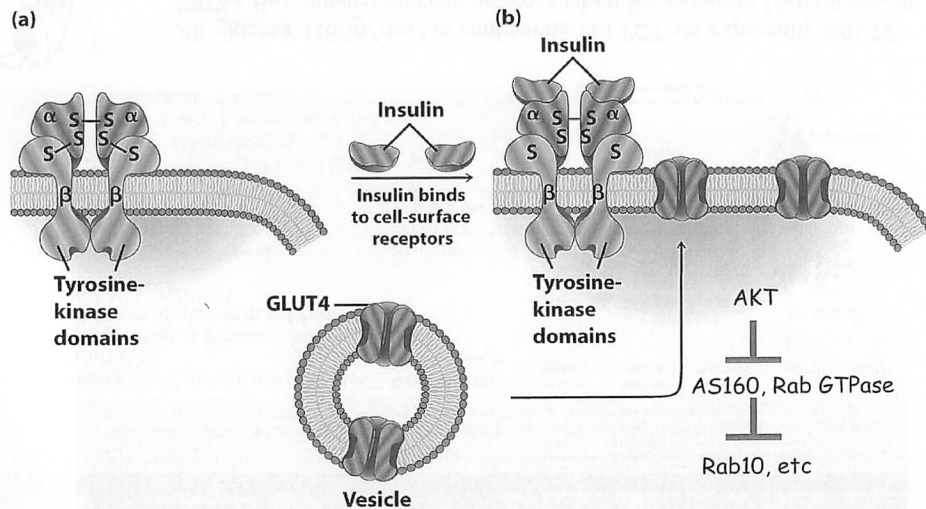


Figure 11-13 Principles of Biochemistry, 4/e
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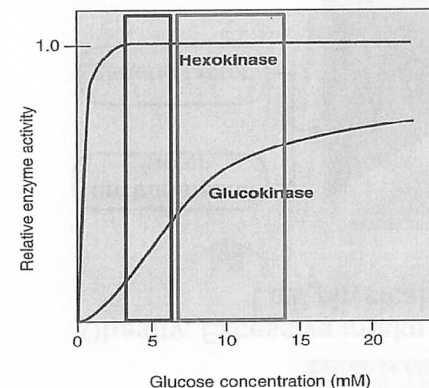
In skeletal muscle, contraction also stimulates GLUT4 translocation via Ca^{2+} /calmodulin and AMPK

Hexokinase and GLUT

HKII and **GLUT4** are colocalized primary in insulin-sensitive tissues such as muscle and adipose tissues.

GK and **GLUT2** are simultaneously expressed in liver.

Q: Explain effects of concomitant expressions of these HK and GLUT on the regulation of blood sugar levels

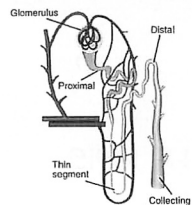


Name	Tissue location	K_m	Comments
GLUT2	Liver and pancreatic β cells	15–20 mM	In the pancreas, plays a role in regulation of insulin In the liver, removes excess glucose from the blood
GLUT4	Muscle and fat cells	5 mM	Amount in muscle plasma membrane increases with endurance training

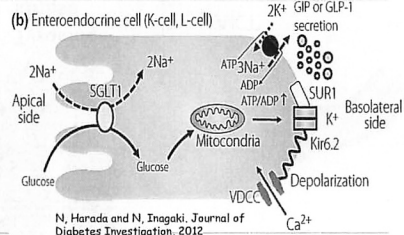
6. SGLT2 (sodium glucose cotransporter 2) inhibitor: Phlorizin analogs

SGLT (sodium myo-inositol cotransporter)

	SGLT1	SGLT2	SGLT3	SGLT4	SGLT5	SGLT6	SGLT7
Distribution	Intestine, kidney Trachea, heart, brain, etc	Kidney Liver, muscle, heart	Intestine Brain Lung	Intestine Kidney Liver Brain	Kidney	Kidney Brain	
Functions	Absorption of glucose and galactose in intestine, Reabsorption of glucose in kidney, Transport of glucose in sick heart (diabetes, ischemia)	Reabsorption of Glucose in Kidney	Glucose sensor?	mannose fructose?	mannose fructose?	myo-inositol glucose	myo-inositol
Disease due to deficiency	- Lethal diarrhea due to malabsorption of glucose and galactose - In DM, its activity increased (3X) - Insulin stimulate its activity by recruiting SGLT1 to membrane in PKC-dependent manner	Glycosuria Increased expression in DM					



- All glucose** (160g/day) is reabsorbed at PCT via especially **SGLT2** (90%). But urinary glucose appears when BS exceeds 160-180mg/dl.
- SGLT2 is present predominantly in early (S1 segment) and late (S2 segment) PCT, whereas SGLT1 is located in late (S3 segment) PCT.



N. Harada and N. Inagaki. Journal of Diabetes Investigation, 2012

Diabetes mellitus

	Type I (5-10 %)	Type II
Etiology	Autoimmune destruction of β cells	Insulin resistance with inadequate β cell function to compensate
Insulin levels	Absent or negligible	Typically higher than normal
Insulin action	Absent or negligible	Decreased
Insulin resistance	Not part of syndrome, but may be present	Yes
Age of onset	Typically ≤ 30 years	Typically ≥ 40 years
Acute complications	Ketoacidosis* , wasting	Hyperglycemia (can lead to hyperosmotic seizure and coma)
Chronic complications	Micro- and Macrovascular diseases	Same as Type I
Pharmacological interventions	insulin	A number of drugs are available, including insulin if others fail

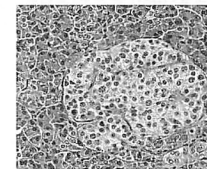
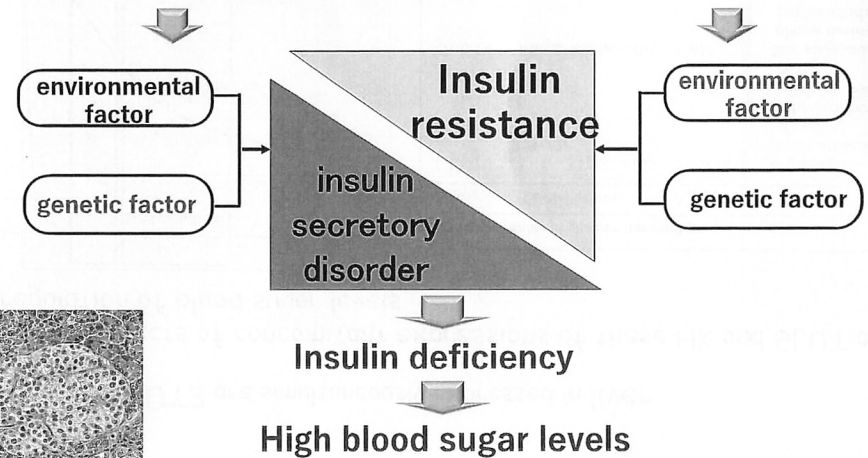
*Diabetic Ketoacidosis, a unique feature of Type I DM, occurs by increasing ketone body due to increased influx of FFA from adipose tissues where uncontrolled lipolysis is present and resultant decreasing bicarbonate.

- ✓ DM caused by other diseases such as genetic disorders, pancreatic diseases, liver diseases, endocrine abnormality, drugs
- ✓ Gestational diabetes

80% of Type II DM is obesity

Diabetes mellitus (糖尿病)

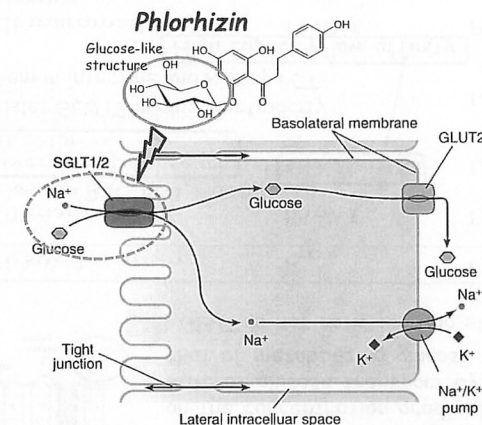
Obesity, Excessive intake of high caloric and lipid-rich diets,
Low physical activity, Stress, Aging

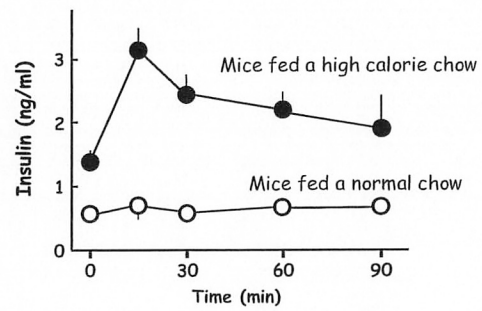
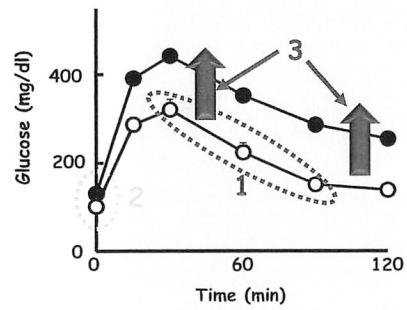


99% of pancreatic mass consist of exocrine cell, while only one million small islands. Islet of Langerhans are endocrine tissues, and consist of four distinct cell type, α cells (glucagon), β cells (insulin), δ cells (somatostatin and gastrin), and PP cells (pancreatic polypeptide).

In diabetic patients, renal reabsorption of glucose is **enhanced** compared to healthy human !!

➤ Phlorizin analogs are **competitive** SGLT2 inhibitors





Q1: Blood glucose levels decrease 30 mins after glucose challenge.

1. What organs/cells predominantly contribute to this decrease?
2. What mechanisms exert to decrease blood glucose levels?

Q2: Blood glucose levels increase even in 24hr-fasting conditions.

1. What organs/cells predominantly contribute to this increase?
2. What mechanisms exert to increase blood glucose levels?

Q3: Blood glucose levels further increase after glucose challenge.

1. What organs/cells predominantly contribute to this increase?
2. What mechanisms exert to increase blood glucose levels?